1. What does TDM stand for?

A) Total Drug Management

B) Therapeutic Drug Monitoring

C) Toxic Drug Measurement

D) Targeted Dose Modulation

2. The main goal of TDM is to:

A) Reduce cost of therapy

B) Maintain drug concentration within therapeutic range to maximize efficacy and minimize toxicity

C) Increase drug half life

D) Avoid sample collection

3. Which is the correct order of TDM Cycle?

A) Analysis → Collection → Judgement → Dosage → Personalized Care

B) Collection → Analysis → Interpretation → Medical Judgement → Dosage Adjustment → Personalized Care

C) Collection → Judgement → Analysis → Dosage → Interpretation → Care

D) Dosage → Collection → Analysis → Care

4. Why is timing of sample collection critical?

A) To save laboratory time

B) Because drug concentration changes with time after dose, wrong timing leads to misinterpretation

C) Because patient prefers it

D) Timing is not important

5. Trough level is defined as:

A) Highest concentration after a dose

B) Lowest concentration just before next dose

C) Average concentration during day

D) Concentration in urine

6. Which analytical technique is considered gold standard for specificity in TDM?

A) Immunoassays

B) LC-MS/MS (Liquid Chromatography-Tandem Mass Spectrometry)

C) Colorimetry

D) Dipstick

7. Immunoassays are preferred when:

A) High specificity is needed for metabolites

B) Rapid, automated routine measurement is needed

C) No lab equipment is available

D) Drug has no antibodies

8. What does Subtherapeutic mean in Interpretation step?

A) Within desired range

B) Above toxic level

C) Below effective level, may cause treatment failure

D) Equal to toxic level

9. Supratherapeutic level indicates:

A) Treatment failure

B) Increased risk of toxicity

C) Need to increase dose

D) Perfect dose

10. Medical Judgement step includes all EXCEPT:

A) Age and weight

B) Liver and kidney function

C) Patient's favorite color

D) Concomitant drugs and clinical response

11. If vancomycin trough is too high and creatinine is rising, the appropriate action is:

A) Increase dose

B) Decrease dose or extend interval and reassess renal function

C) Continue same dose

D) Switch to oral form

12. Which patient factor most affects drug clearance and needs consideration in dose adjustment?

A) Eye color

B) Renal and hepatic function

C) Height alone

D) Blood group

13. Personalized Care in TDM means:

A) Same dose for all patients

B) Tailoring therapy to individual patient characteristics and drug levels

C) Using only branded drugs

D) Avoiding follow up

14. TDM is MOST needed for drugs with:

A) Wide therapeutic index and no toxicity

B) Narrow therapeutic index, high inter-individual variability, severe toxicity

C) Topical action only

D) No systemic absorption

15. Which of the following requires routine TDM?

A) Paracetamol in normal dose

B) Tacrolimus after organ transplant

C) Vitamin C

D) Oral antacid

16. What happens after dosage adjustment in TDM Cycle?

A) Cycle ends permanently

B) Cycle restarts with new sample collection to verify the change

C) Patient is discharged without follow up

D) No need for further monitoring

17. Biosensors in Sample Analysis offer:

A) Slower results than lab

B) Potential for point of care and real time monitoring

C) No clinical use

D) Only measure proteins

18. A high phenytoin level with nystagmus and ataxia suggests:

A) Subtherapeutic level

B) Therapeutic level

C) Toxic level requiring dose reduction and clinical evaluation

D) Need to increase dose

19. Drug interaction assessment is important in TDM because:

A) Interacting drugs can alter drug metabolism and concentration

B) It has no effect on TDM

C) Only food affects TDM

D) Interactions only affect taste

20. Which statement best describes TDM as a cycle?

A) It is a linear one time process

B) It is a continuous feedback loop: measure, interpret, decide, adjust, re-measure for optimal outcome

C) It only involves laboratory

D) It replaces clinical judgement completely